

Does the IVR Signal Survive Full Momentum Reconstruction?

A detector-level y-pol closure study with ^{112}Sn and ^{124}Sn
190 MeV/u with SAMURAI + NEBULA Plus

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What Are We Trying to Establish?

Physics question

Can the isovector reorientation (IVR) signal from a γ -polarized deuteron remain observable after the realistic SAMURAI detector and momentum-reconstruction chain?

Why two isotopes?

^{112}Sn provides an isotope reference.

^{124}Sn provides the stronger neutron-rich response.

Current answer

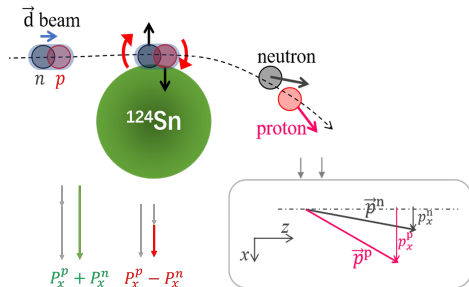
For ^{124}Sn , the γ -dependent ordering survives reconstructed proton, neutron, and event-plane determination in a controlled neutron fiducial region.

Scope

The observable and px60 fiducial are reconstructed; the present loose/mid/tight event-quality class is still defined with truth momenta.

This is a detector-level closure study, not yet a complete experimental uncertainty analysis.

Physical Picture: Isovector Reorientation



- A polarized deuteron approaches a neutron-rich target.
- Proton and neutron experience different isovector mean-field forces.
- The breakup fragments acquire a polarization-dependent relative momentum bias.

$$E_{\text{sym}}(\rho) = E_{\text{sym}}^{\text{kin}}(\rho) + C \left(\frac{\rho}{\rho_0} \right)^\gamma$$

Model parameter

γ parametrizes the density dependence used in the ImQMD calculation.

Why Compare Two Tin Isotopes?

target	Z	N	role
^{112}Sn	50	62	isotope reference
^{124}Sn	50	74	primary sensitivity

- Same proton number controls much of the target charge and kinematic environment.
- Different neutron excess changes the isovector mean-field response.
- The same detector/reconstruction chain is required for a controlled isotope comparison.

Takeaway

A common detector effect should influence both isotopes, while an isovector response should change with target neutron excess. ^{124}Sn is the primary sensitivity target; ^{112}Sn is the isotope reference.

Why Start with y Polarization?

- y-pol and z-pol are both physically valuable IVR channels.
- y-pol currently has the most mature combined simulation, detector folding, momentum reconstruction, and analysis baseline.
- y-pol also has an established left-right-up-down (LRUD) beam-monitoring strategy for vertical tensor polarization p_{yy} .

First-stage baseline

y-pol provides a realistic first-stage path to testing the IVR concept at SAMURAI.

z-pol

z-pol remains a second-stage extension after successful commissioning.

Reconstructed IVR Observable

$$\phi_{\text{event}}^{\text{reco}} = \text{atan2}(p_{y,p}^{\text{reco}} + p_{y,n}^{\text{reco}}, p_{x,p}^{\text{reco}} + p_{x,n}^{\text{reco}})$$

$$\Delta p_{x,\text{reco}}^{\text{rot}} = p_{x,p,\text{reco}}^{\text{rot}} - p_{x,n,\text{reco}}^{\text{rot}}$$

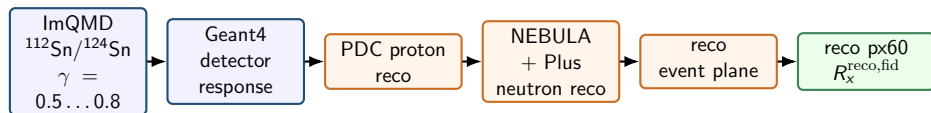
$$R_x^{\text{reco,fid}} = \frac{N(\Delta p_{x,\text{reco}}^{\text{rot}} > 0)}{N(\Delta p_{x,\text{reco}}^{\text{rot}} < 0)}$$

Experimental comparison

$$R_{x,\text{data}}^{\text{reco,fid}} \quad \text{vs.} \quad R_{x,\text{model}}^{\text{fold,fid}}(\gamma)$$

This is a reconstructed fiducial observable, not a global truth-level ratio.

From ImQMD Events to the Measured Observable



$$N_{\text{model, reco}, j}(\gamma) = \sum_i K_{ji} N_{\text{model, true}, i}(\gamma) + b_j$$

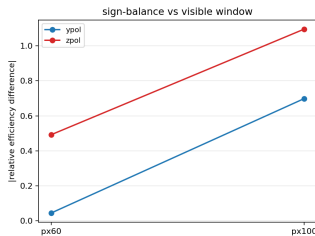
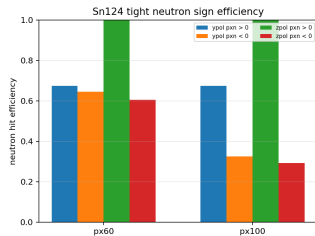
Current closure

reco observable + reco px60 + truth-defined quality class

Final experimental analysis

reco observable + reco px60 + reco-defined quality class

Defining a Controlled Neutron Fiducial Region



Neutron acceptance is strongly phase-space dependent; full phase space contains near-zero-efficiency regions.

$$\epsilon_+ = 0.678, \quad \epsilon_- = 0.644$$

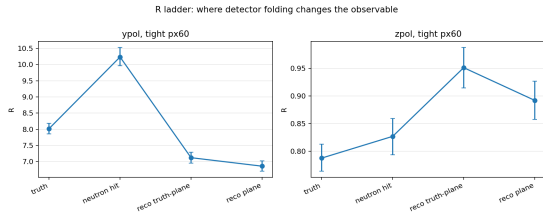
These are neutron-hit efficiencies in the truth-side $|p_{x,n}| < 60$ MeV/c fiducial, pooled over y-pol tight events and all γ samples.

px60

px60 means the reconstructed neutron fiducial $|p_{x,n}^{\text{reco}}| < 60$ MeV/c.

Takeaway: px60 defines a fiducial detector-level measurement. It does not reconstruct the unobserved full phase space.

Where Does Detector Folding Change the Observable?

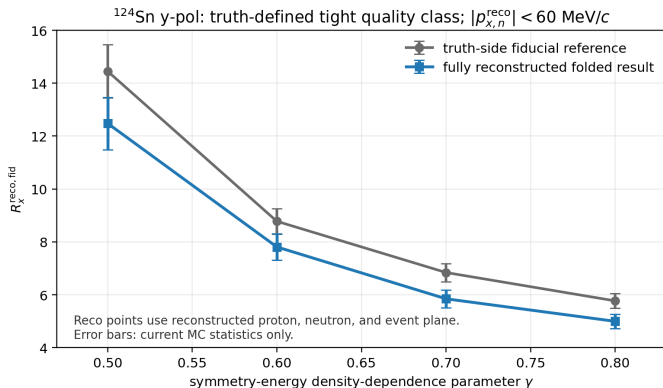


- 1 Neutron visibility/acceptance produces the largest shift.
- 2 Proton momentum reconstruction is not the dominant bottleneck.
- 3 Event-plane reconstruction adds a smaller correction.

Takeaway

The absolute ratio changes, but the γ ordering remains.

Primary Sensitivity: ^{124}Sn

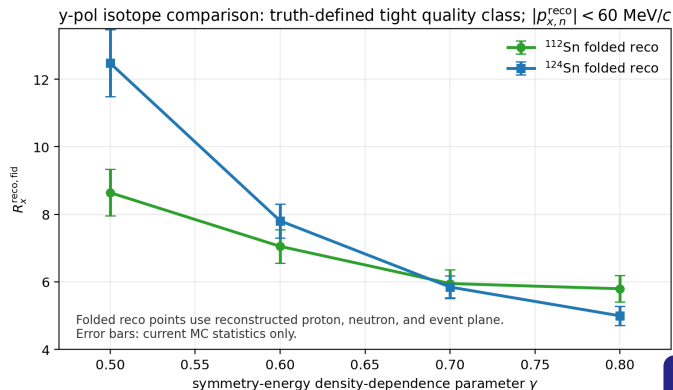


Current closure selection: truth-defined tight event-quality class + reconstructed px60 fiducial.

Takeaway

For ^{124}Sn , the monotonic γ ordering survives the complete proton, neutron, and event-plane reconstruction chain. Error bars are current MC statistics only.

Isotope Reference: ^{112}Sn versus ^{124}Sn



- ^{124}Sn remains the primary sensitivity target.
- ^{112}Sn now has the same tight-px60 folded-reco table.
- Both targets keep a visible y-pol γ ordering after folding.
- The high- γ Sn112 interval is narrower and more systematics sensitive.

Scope

Reconstructed observable and reconstructed px60 fiducial; tight event-quality class is still truth-defined. Sn112 planning rates use an atom-density scaling placeholder.

Statistical Scale versus Experimental Systematics

Closest adjacent interval in the current ^{124}Sn y-pol folded result:

$$\gamma = 0.7 \rightarrow 0.8, \quad N_{\text{stat},3\sigma} \simeq 2.8 \times 10^3.$$

Planning scale

Current planning anchor gives about 2.75×10^5 usable y-pol events for ^{124}Sn .

Not included

Detector, polarization, background, and model uncertainties are not included in this statistical-only estimate.

Systematic priorities

neutron efficiency; threshold and time-of-flight T_0 ; cross-talk and fake neutrons; backgrounds; p_{yy} polarization; detector alignment; model dependence; reconstructed quality-class migration.

Current Status and Next Closure Step

Established

- y-pol IVR observable is reconstructable.
- ^{124}Sn γ ordering survives detector folding.
- px60 provides a controlled fiducial region.
- folded model comparison is the right strategy.

Still required

- reco-defined event-quality classes,
- reco-defined Sn112/Sn124 isotope closure,
- Sn112 target-specific rate validation,
- selection efficiency, purity, and migration,
- polarization and detector-systematic pseudo-data tests.

The next decisive analysis is not another truth-level calculation. It is a closure test with reconstructed quality-class definitions.

Summary

- ① IVR produces a polarization-dependent proton-neutron relative-momentum bias near a neutron-rich target.
- ② ^{124}Sn is the primary sensitivity target, while ^{112}Sn provides a controlled isotope reference.
- ③ The γ -dependent ordering survives realistic detector folding and full proton, neutron, and event-plane reconstruction in the px60 fiducial region for ^{124}Sn .
- ④ The experiment should compare reconstructed data with identically folded model predictions; the next step is closure with reconstructed quality classes and detector-systematic validation.

Bottom line

The present study establishes detector-level observability, not yet a complete experimental uncertainty budget. A y-pol measurement provides a realistic first-stage path to testing the IVR concept at SAMURAI.

Thanks

Questions?

Backup Roadmap

- A: model and observable
- B: reconstruction
- C: neutron acceptance
- D: ^{124}Sn full numbers
- E: ^{112}Sn and isotope comparison
- F: z-pol extension
- G: polarization monitoring
- H: systematics and future closure

Backup A: Observable and Bounded Asymmetry

$$\Delta p_{x,\text{reco}}^{\text{rot}} = p_{x,p,\text{reco}}^{\text{rot}} - p_{x,n,\text{reco}}^{\text{rot}}, \quad R_x^{\text{reco,fid}} = \frac{N(\Delta p_{x,\text{reco}}^{\text{rot}} > 0)}{N(\Delta p_{x,\text{reco}}^{\text{rot}} < 0)}.$$

$$A_x = \frac{N_+ - N_-}{N_+ + N_-} = \frac{R_x - 1}{R_x + 1}.$$

Vocabulary

Truth, reconstructed, folded, and unfolded observables are different objects. The main result uses a reconstructed fiducial observable.

Backup A: Folding Vocabulary

$$N_{\text{reco},j} = \sum_i K_{ji} N_{\text{true},i} + b_j.$$

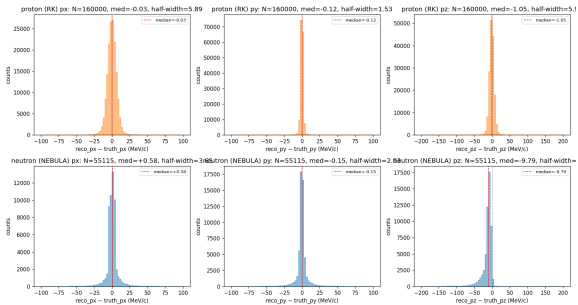
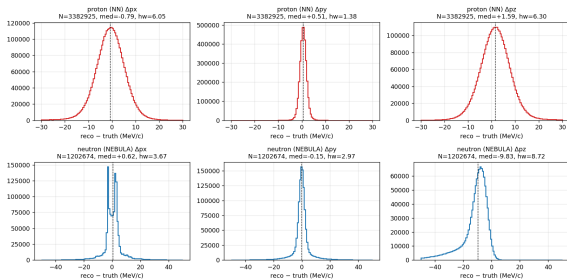
- truth: generator-level kinematics after physics selection,
- reconstructed: variables built from detector hits,
- folded: model events passed through detector response and reconstruction,
- unfolded: estimator of truth after response inversion and regularization.

Current limitation

The observable numerator/denominator and neutron px60 fiducial are reconstructed. The loose/mid/tight event-quality class is still defined with truth momenta in the current closure.

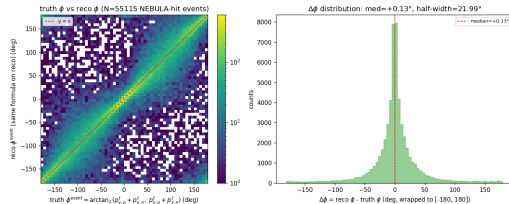
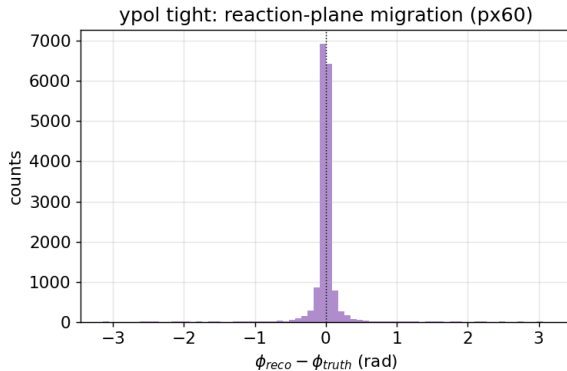
Backup B: Proton and Neutron Residuals

y-pol: per-particle residuals (NN proton + NEBULA neutron)



Left: NN breakup diagnostic. Right: earlier RK+NEBULA y-pol full-reco study. Half-width and RMSE should not be mixed; tails matter for RMSE.

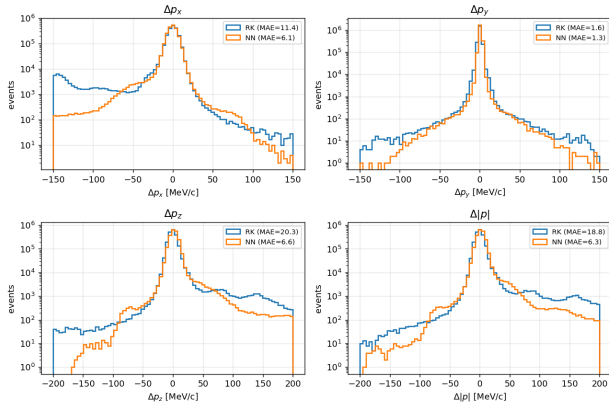
Backup B: Event-Plane Resolution



Reaction-plane migration broadens the observable but is not the leading change in the R ladder.

Backup B: RK vs NN Proton Cross-Check

Proton residuals — ypol



Strict RK–NN comparison is available on the common configuration without NEBULA Plus.

y-pol metric	RK	NN
Δp_x half-width	5.86	6.01
Δp_x RMSE	39.7	13.3
Δp_z half-width	5.88	6.25
Δp_z RMSE	87.8	15.7

Proton reconstruction is cross-checked and is not the dominant observable shift.

Backup B: NN Proton Backend

Inputs

PDC1/PDC2 hit positions:

$$(x_1, y_1, z_1, x_2, y_2, z_2).$$

Outputs

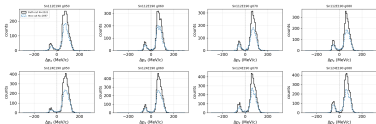
Target-frame proton momentum:

$$(p_x, p_y, p_z)_{\text{target}}.$$

- MLP backend used through the canonical reconstruction CLI.
- Training domain covers the breakup kinematics used here.
- NN has comparable core resolution to RK and smaller catastrophic tails in the common-geometry comparison.

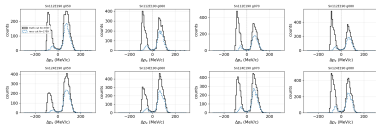
Backup C: px60, px80, px100

ypol Δp_x , truth-cut vs reco-plane reco-cut (tight, px60)



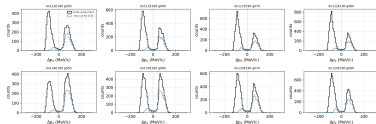
px60

ypol Δp_x , truth-cut vs reco-plane reco-cut (tight, px80)



px80

ypol Δp_x , truth-cut vs reco-plane reco-cut (tight, px100)

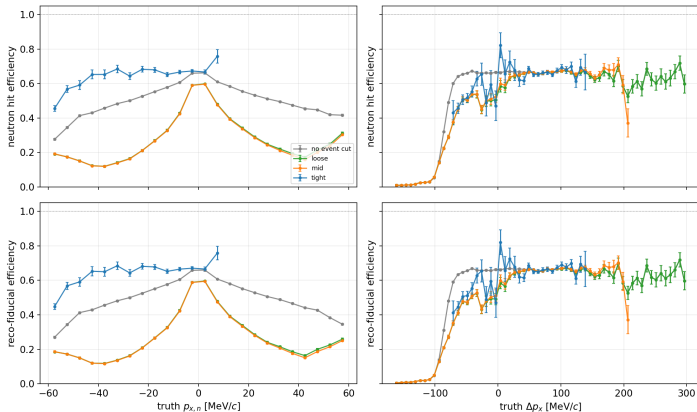


px100

px80 improves over px100, but px60 removes most of the low-efficiency negative- $p_{x,n}$ branch.

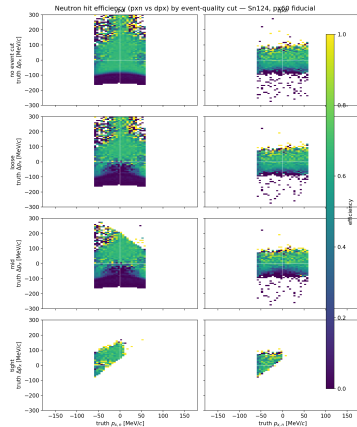
Backup C: 1D Neutron Efficiency

Sn124 ypol: neutron efficiency by event-quality cut (px60 fiducial, bins $N \geq 20$)



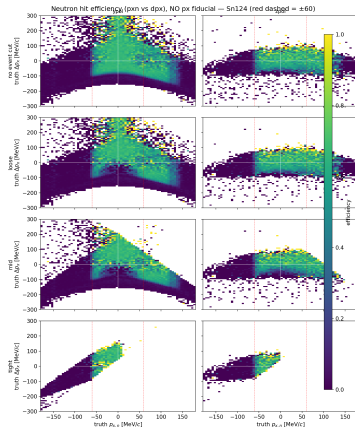
Efficiency shape is similar across event-quality cuts; cuts mainly change which phase-space region is populated.

Backup C: 2D Neutron Efficiency



The response is not uniform in $(p_{x,n}, \Delta p_x)$; this motivates folded model comparison.

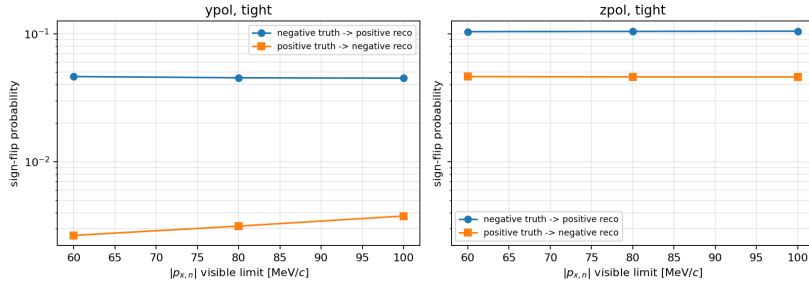
Backup C: No-Fiducial Efficiency Landscape



Near-zero-efficiency regions make a simple $1/\epsilon$ correction or raw unfolding unstable.

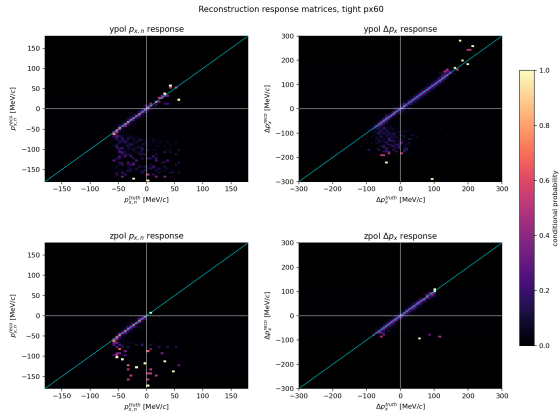
Backup C: Sign Migration

Δp_x sign migration after reconstruction



In y-pol tight p_{x60} , Δp_x sign migration is small compared with the neutron visibility effect.

Backup C: Response Matrices



Response matrices are useful for closure and possible unfolding, but the main result does not rely on matrix inversion.

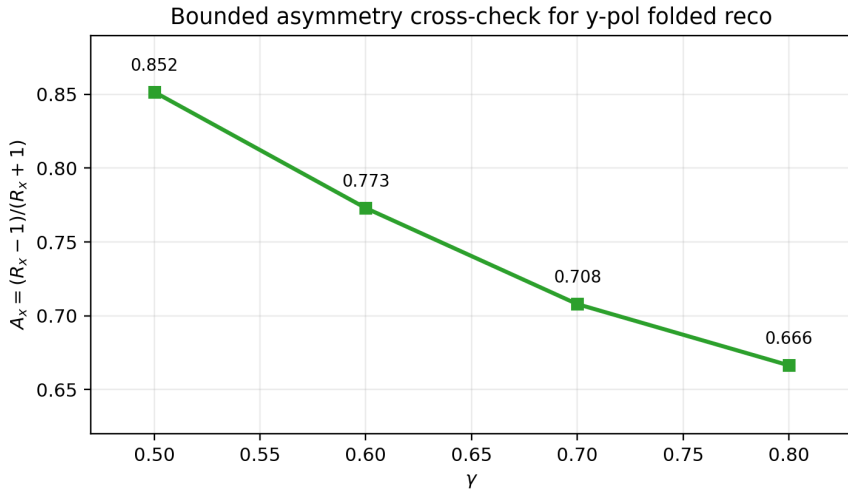
Backup D: ^{124}Sn Full y-pol Numbers

γ	N_+	N_-	$R_x^{\text{reco, fid}} \pm \sigma_{\text{MC}}$	A_x	N_{used}
0.5	2134	171	12.48 ± 0.99	0.852	2305
0.6	2147	275	7.81 ± 0.50	0.773	2422
0.7	2093	358	5.85 ± 0.33	0.708	2451
0.8	1948	390	4.99 ± 0.28	0.666	2338

Meaning

These are folded reco-plane values for ^{124}Sn y-pol tight px60, not truth-level asymmetries. Uncertainties are current MC statistics only.

Backup D: Bounded Asymmetry Cross-Check



A_x is bounded and monotonic for the same reconstructed y-pol points. It is a stability cross-check, not a replacement for the primary R_x result.

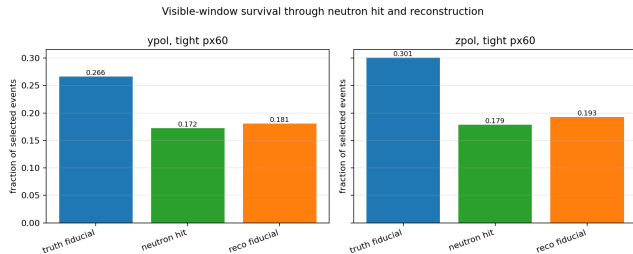
Backup D: Adjacent Gamma Separation

interval	R_{left}	R_{right}	$ \Delta R $	$N_{\text{stat},3\sigma}$
0.5→0.6	12.48	7.81	4.67	519
0.6→0.7	7.81	5.85	1.96	979
0.7→0.8	5.85	4.99	0.85	2.77×10^3

Scope

This is a statistical-only planning scale, not an experimental significance claim.

Backup D: Usable-Event Survival



Planning formula:

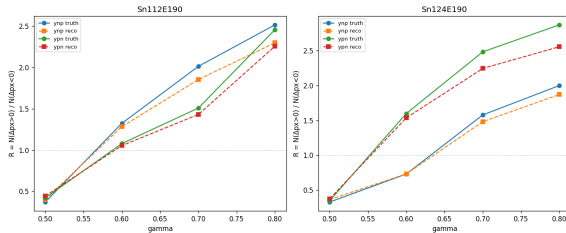
$$N_{\text{usable}} = \text{rate} \times T \times \frac{N_{\text{reco plane}}}{N_{\text{raw}}}.$$

For y-pol:

$$\frac{N_{\text{reco plane}}}{N_{\text{raw}}} = 0.00566$$

$$N_{\text{usable}} \simeq 2.75 \times 10^5.$$

Backup E: Earlier ^{112}Sn Physics-Motivation Result



This earlier NN-breakup result already showed Sn112/Sn124 reconstruction behavior before the current NEBULA Plus px60 gamma-constraint tables were exported.

Use in this talk

It remains useful as a reconstruction-history and physics-motivation cross-check. The main isotope statement now uses the current tight-px60 folded-reco table.

Backup E: Sn112/Sn124 Common-Chain Audit

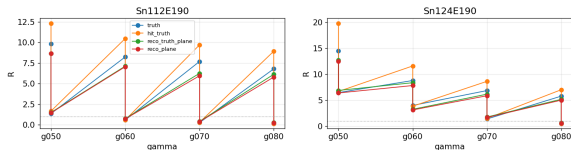
item	Sn112/Sn124 status
production	NEBULA Plus joint 20260609 tables exported for both targets
geometry/backend	SAMURAI + NEBULA Plus, PDC NN proton, neutron time-of-flight
y-pol γ grid	0.5, 0.6, 0.7, 0.8 for both targets
observable	folded reco-plane $R_x^{\text{reco, fid}}$ for both targets
selection	truth-defined tight quality class + reco px60 fiducial
remaining caveat	reco-defined quality class and Sn112 target-specific rate still pending

Decision

A direct y-pol folded-reco isotope plot is now appropriate, with the truth-defined quality-class and planning-rate caveats shown explicitly.

Backup F: z-pol Folded Results

zpol tight: R ladder vs gamma (px60)



γ	N_+	N_-	R_{reco}
0.5	306	48	6.38
0.6	208	67	3.10
0.7	214	129	1.66
0.8	171	307	0.56

z-pol remains physically valuable but is kept as a second-stage extension.

Backup F: y-pol and z-pol Cut Definitions

y-pol tight

$$|p_{y,p}^{\text{truth}} - p_{y,n}^{\text{truth}}| < 150$$

$$50 < |\vec{p}_{T,p}^{\text{truth}} + \vec{p}_{T,n}^{\text{truth}}| < 200$$

$$\pi - |\phi_{\text{truth}}| < 0.2.$$

z-pol tight

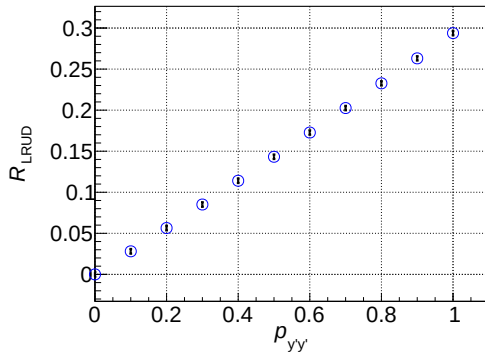
$$p_{z,p}^{\text{truth}} + p_{z,n}^{\text{truth}} > 1150$$

$$|p_{z,p}^{\text{truth}} - p_{z,n}^{\text{truth}}| < 150$$

$$|\vec{p}_T^{\text{sum}}| > 50, \quad \pi - |\phi_{\text{truth}}| < 0.5.$$

These current event-quality classes are defined with truth momenta and must be replaced by reconstructed definitions.

Backup G: y-pol Polarization Monitoring

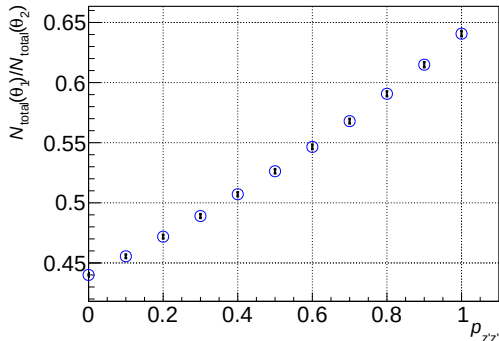


Vertical tensor polarization p_{yy} can be monitored through an established LRUD asymmetry in deuteron-proton elastic scattering.

Role in IVR

Polarization monitoring is an independent prerequisite and must be propagated into $R_x(\gamma)$.

Backup G: z-pol Monitoring Concept



Longitudinal tensor polarization p_{zz} can be monitored with a more demanding absolute/two-angle normalization strategy.

Careful statement

p_{zz} monitoring is not impossible; it is more complex than the y-pol LRUD baseline.

Backup H: Truth-Assisted Cuts

What is fully reconstructed now

Numerator and denominator of $R_x^{\text{reco, fid}}$ use reconstructed proton momentum, reconstructed neutron momentum, and reconstructed event plane.

What still uses truth

Event-quality classes such as transverse balance and near-backscatter selection are currently defined using truth momenta in simulation. The px60 fiducial used for the folded observable is already reconstructed.

Required closure

Re-run with reconstructed quality-class definitions and compare efficiency, purity, migration, and γ ordering against the truth-defined baseline.

Backup H: Detector-Systematic Checklist

- neutron threshold, T_0 , timing resolution, and hit-position model,
- cross-talk, fake neutron candidates, and accidental coincidences,
- charge-veto and charged-particle leakage assumptions,
- detector geometry and alignment variations,
- background and pile-up in the relevant electronics time window,
- polarization uncertainty and model/folding uncertainty.

Next analysis product

Pseudo-data closure under detector and model variations, with the same folded comparison used for the experimental observable.